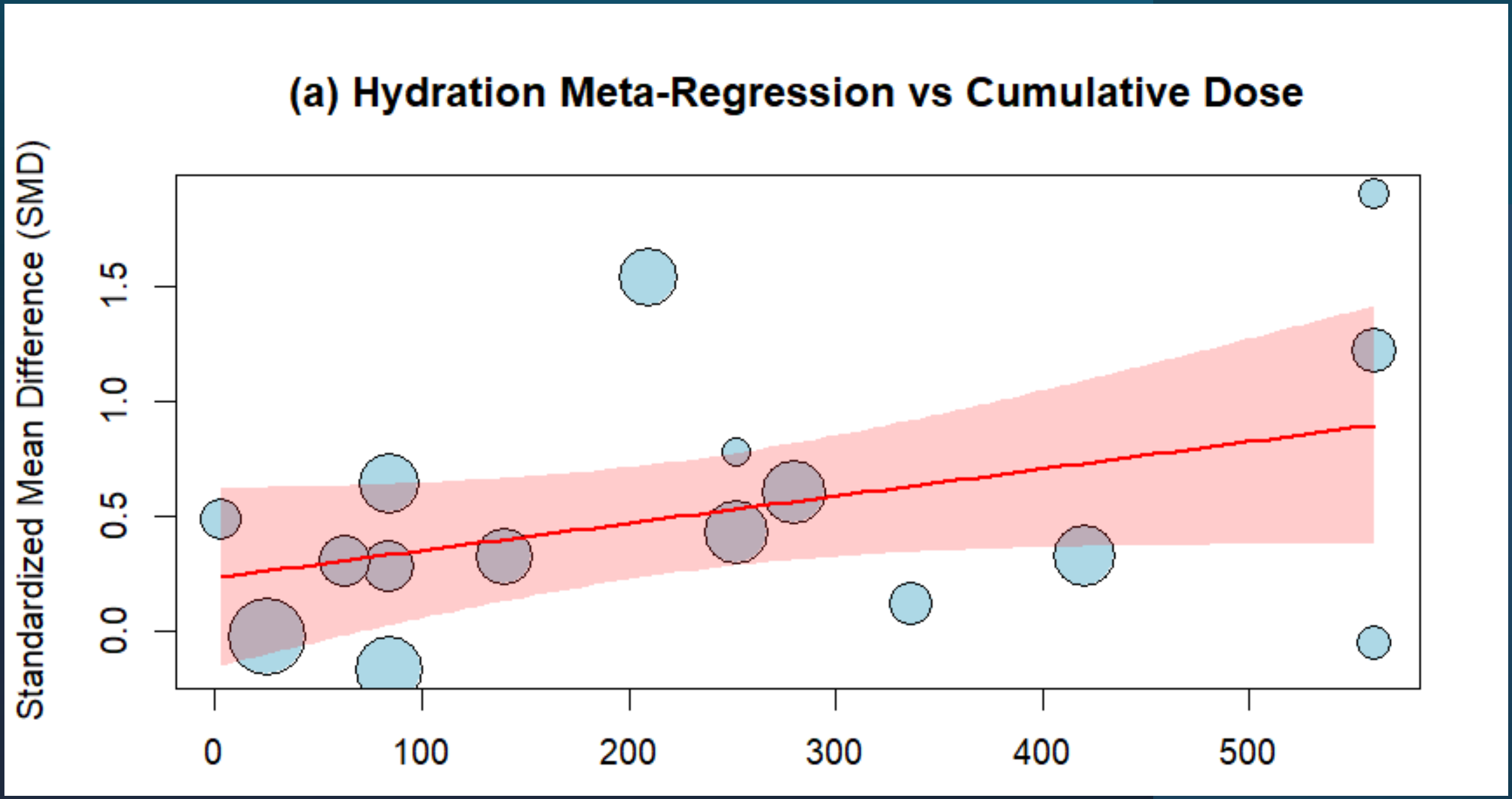
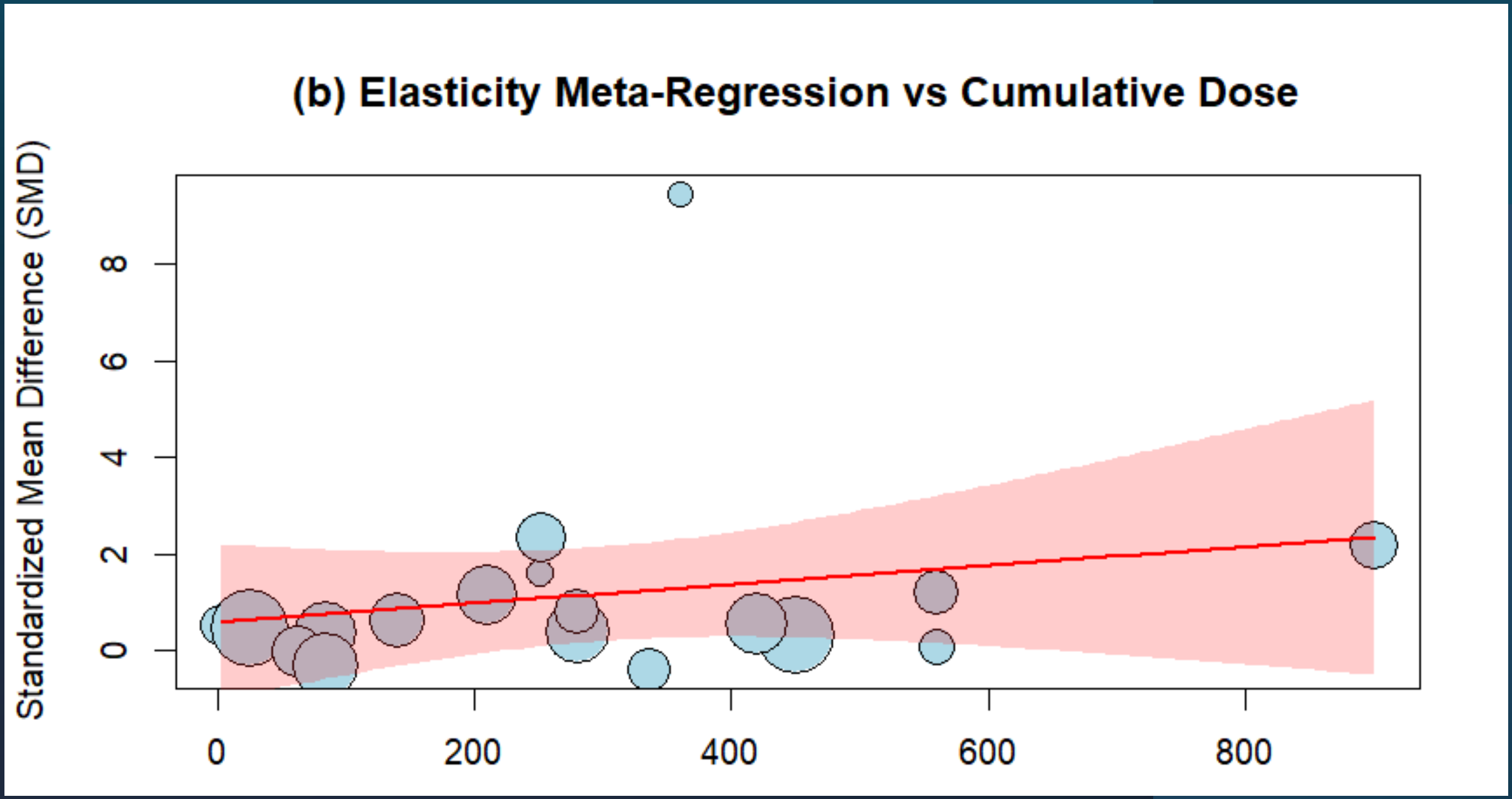

(a) Hydration Meta-Regression vs Cumulative Dose
Standardized Mean Difference (SMD)
0.0
0.5
1.0
1.5
0
100
200
300
400
500


(b) Elasticity Meta-Regression vs Cumulative Dose
Standardized Mean Difference (SMD)
0
2
4
6
8
0
200
400
600
800

Hydration meta-regression results

Note that none of these regressions are truly meaningful, even if data was reliable, as they use SMD from heterogeneous methods, . The illustration is that dose must have a biological rationale BEFORE analysis is run – otherwise analysis risks being post-hoc data-dredging and non-informative.

Dose regression

```
Model Results:

         estimate      se    zval    pval    ci.lb   ci.ub
intrcpt    0.2574  0.1794  1.4348  0.1513  -0.0942  0.6089
dose       0.0716  0.0398  1.8000  0.0719  -0.0064  0.1495   .

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Dose and time regression

```
Model Results:

         estimate      se     zval    pval    ci.lb   ci.ub
intrcpt    0.4098  0.5814   0.7049  0.4809  -0.7297  1.5493
dose       0.0681  0.0427   1.5966  0.1104  -0.0155  0.1518
weeks     -0.0136  0.0494  -0.2741  0.7840  -0.1104  0.0833

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Cumulative dose regression

```
Model Results:

            estimate      se    zval    pval    ci.lb   ci.ub
intrcpt       0.2296  0.1987  1.1556  0.2478  -0.1598  0.6190
total_dose    0.0012  0.0007  1.7022  0.0887  -0.0002  0.0026   .

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Elasticity meta-regression results

Note that none of these regressions are truly meaningful, even if data was reliable, as they use SMD from heterogeneous methods, . The illustration is that dose must have a biological rationale BEFORE analysis is run – otherwise analysis risks being post-hoc data-dredging and non-informative.

Dose regression

```
Model Results:

         estimate      se    zval    pval    ci.lb   ci.ub
intrcpt    0.8853  0.8253  1.0726  0.2834  -0.7323  2.5029
dose       0.0691  0.1626  0.4250  0.6708  -0.2496  0.3878

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Dose and time regression

```
Model Results:

         estimate      se    zval    pval    ci.lb   ci.ub
intrcpt   -0.9815  2.1102 -0.4651  0.6418  -5.1174  3.1544
dose       0.0752  0.1636  0.4595  0.6459  -0.2454  0.3958
weeks      0.1759  0.1828  0.9623  0.3359  -0.1824  0.5342

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Cumulative dose regression

```
Model Results:

            estimate      se    zval    pval    ci.lb   ci.ub
intrcpt       0.5943  0.8200  0.7248  0.4686  -1.0129  2.2016
total_dose    0.0019  0.0022  0.8695  0.3846  -0.0024  0.0063

---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```